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RESEARCH ARTICLE

Lecanemab treatment for Alzheimer's Disease of varying severities and associated plasma biomarkers monitoring: A multi-center real-world study in China

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Abstract

INTRODUCTION: We investigated real-world efficacy, safety, and plasma biomarker dynamics of Lecanemab in Chinese patients with Alzheimer's disease (AD).

METHODS: A multi-center prospective cohort study enrolled 68 AD patients. Cognitive scales and plasma biomarkers were assessed at baseline (V0), 2.5 months (V1), and 7 months (V2).

RESULTS: Alzheimer's Disease Assessment Scale-Cognitive Subscale 14-item version (ADAS-cog14) scores improved significantly at both follow-ups, and plasma p-tau181 consistently declined. Both p-tau181 and p-tau217 correlated with cognition and partially predicted treatment response (area under the curve [AUC] = 0.734 and 0.713). Mixed-effects modeling confirmed their dynamic association with ADAS-cog14 scores. Subgroup analyses indicated benefits across sex and *apolipoprotein E4* status, while moderate-to-severe cases showed limited response. Lecanemab was well tolerated, with asymptomatic amyloid-related imaging abnormalities in 17.65% and mild infusion reactions in 5.88%.

Sihui Chen, Ruwei Ou, Xueping Chen, and Huifang Shang contributed equally to this study.

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DISCUSSION: These findings support the short-term efficacy and safety of Lecanemab in early AD and highlight plasma biomarkers as a treatment-responsive biomarker.

KEYWORDS
Alzheimer's disease, lecanemab, plasma biomarkers, real-world study

Highlights

- Lecanemab improved cognitive function in Chinese patients with mild cognitive impairment due to Alzheimer's disease (AD-MCI) and mild AD over a short period.
- Plasma p-tau181 and p-tau217 showed significant correlation with cognitive scores, and their baseline level could partially predict the efficacy of lecanemab.
- Lecanemab showed a favorable safety profile with low, manageable rates of amyloid-related imaging abnormalities (ARIA) and infusion reactions.

1 | BACKGROUND

Alzheimer's disease (AD) is the leading cause of dementia and is rapidly becoming one of the most costly, deadly, and burdensome diseases of this century.[1,2] In the United States, over 6.9 million people aged 65 and older are living with AD, and this number is projected to increase to 13.8 million by 2060.[3] In China, it has the largest population of individuals with dementia in the world, AD ranks as the fifth leading cause of death in the country.[4] Mild cognitive impairment (MCI) due to AD (AD-MCI) and mild AD represent early stages of the disease continuum, with a global MCI prevalence of approximately 15% among individuals over 50 years of age.[5–9] Furthemore, a significant proportion of AD-MCI individuals are likely to progress to dementia, imposing substantial burdens on families, healthcare infrastructures, and society.[10,11]

Lecanemab, a humanized immunoglobulin (Ig) G1 monoclonal antibody targeting soluble Aβ protofibrils with high affinity, has demonstrated efficacy in reducing amyloid biomarkers and slowing cognitive decline in patients with AD-MCI and mild AD in clinical trials.[12,13] Notably, Asian patients in the Phase III Clarity-AD trial exhibited better tolerability and therapeutic response compared to the overall study population, suggesting potential regional or ethnic variability in treatment outcomes.[14] However, pivotal clinical trials did not specifically include participants from China—the country with the largest AD population worldwide. On July 6, 2023, lecanemab received full regulatory approval from the United States Food and Drug Administration (FDA) for the treatment of early AD.[15] To date, only a limited number of real-world studies have evaluated lecanemab treatment in patients with AD-MCI and mild AD, most of which are regionally restricted.[16–21] No comparable real-world evidence has been reported from China.

Common cognitive assessments in AD, such as the Alzheimer's Disease Assessment Scale–Cognitive Subscale 14 (ADAS-cog14), Clinical Dementia Rating-Sum of Boxes (CDR-SB), Alzheimer's Disease Composite Score (ADCOMS), and Alzheimer's Disease Cooperative Study-Activities of Daily Living for Mild Cognitive Impairment (ADCS-ADL-MCI),[22–25] are time-consuming, susceptible to rater bias, dependent on patient cooperation, and often lack sensitivity to subtle cognitive changes.[26] Therefore, the integration of plasma biomarkers is crucial for dynamic monitoring in the context of disease-modifying therapies (DMTs), as they can partially compensate for the limitations of these cognitive assessments.[27] Plasma biomarkers such as Aβ1-42/Aβ1-40, phosphorylated tau 181 (p-tau181), p-tau217, glial fibrillary acidic protein (GFAP) and neurofilament light chain (NfL) provide noninvasive insights into AD pathology and enable early detection of treatment response and disease progression.[12,22] However, it remains uncertain whether they can reliably reflect cognitive changes during treatment or accurately predict therapeutic efficacy in the real world.

This study aims to evaluate the real-world effectiveness and safety of lecanemab among AD patients recruited from multiple centers in western China, investigate the influence of sex and apolipoprotein *E4* (*APOE4*) carrier status on treatment response, analyze dynamic changes in plasma biomarkers, and explore their potential for monitoring cognitive changes during therapy.

2 | METHODS

2.1 | Participants

The inclusion criteria were as follows: patients with AD-MCI or clinically diagnosed AD according to the National Institute on Aging-Alzheimer's Association (NIA-AA) criteria, with amyloid pathology confirmed by amyloid positron emission tomography/computed tomography (Aβ-PET/CT) using 18F-florbetapir, and a whole-brain mean standardized uptake value ratio (SUVR) > 1.10 defined as positive.[23,24] All participants were required to exhibit objective impairments in episodic memory. For patients with a CDR score ≥2, their caregivers were fully informed of the potential benefits and risks of treatment. If

they still requested lecanemab therapy after receiving this information, they were included in the study upon signing the appropriate informed consent forms. Exclusion criteria based on magnetic resonance imaging (MRI) included: > 4 cerebral microhemorrhages (≤10 mm), any macrohemorrhage (> 10 mm) and presence of superficial siderosis. Patients with inadequately controlled bleeding disorders (including platelet counts < 50,000/mm^3) or those receiving anticoagulation therapy (e.g., warfarin) with an international normalized ratio (INR) > 1.5 were excluded either. All patients were administered intravenous lecanemab at a standardized dose of 10 mg/kg biweekly, with no prophylactic medication given for infusion-related reactions.

2.2 | Assessment

Patients were assessed using the CDR-SB, ADAS-cog14, Mini-Mental State Examination (MMSE), Montreal Cognitive Assessment (MoCA), and Frontal Assessment Battery (FAB) at baseline (V0), after 2.5 months of treatment (V1), and after 7 months of treatment (V2). A reduction of ≥4 points in the ADAS-cog14 score was considered to represent the minimal clinically important difference (MCID), indicating an improvement that would be perceived as clinically meaningful.[25] To minimize inter-rater variability and potential subjectivity, all cognitive assessments in this study were conducted by the same qualified professional.

2.3 | Measurements

Fasting blood samples were collected at V0, V1, and V2 in ethylenediaminetetraacetic acid (EDTA) -containing tubes and centrifuged at 2000 × g for 10 min at room temperature to separate plasma. All plasma samples were aliquoted and stored at −80°C within 2 h of collection until analysis. Plasma Aβ1-42, Aβ1-40, p-tau181, GFAP, and NfL concentrations were measured for all patients using the fully automated Elf S240 Chemiluminescence Immunoassay platform (Vazyme, Nanjing, China), which was approved by the National Medical Products Administration of China. Plasma p-tau217 levels in EDTA-anticoagulated samples were quantified using the ultrasensitive single-molecule assay (Simoa) technology (Quanterix, Massachusetts, USA) on the fully automated Simoa HD-X platform (GBIO, Hangzhou, China) according to the manufacturer's instruction.[26,27] The ALZpath p-tau217 Advantage Plus assay kit (catalog no. 104570; Quanterix) was used strictly following the protocol. All plasma samples were diluted 1:3 prior to analysis and assayed in a single run using reagents from the same lot. *APOE* genotype (*ε2/ε3/ε4*) was determined using a human *APOE* Gene Genotyping Kit (Memorigen, Xiamen, China). Whole-exome sequencing was performed using high-throughput sequencing technology in patients suspected of carrying pathogenic variants associated with AD. Additionally, follow-up Aβ-PET/CT scans were planned after one-year treatment. The SUVR values for different brain regions and the whole-brain were recorded both pre- and post-treatment.

RESEARCH IN CONTEXT

1. **Systematic review**: A review of pivotal trials and real-world studies on lecanemab revealed a lack of data specific to the Chinese Alzheimer's disease (AD) population. The efficacy and safety in moderate-to-severe AD remain uncertain, and few studies have assessed the sensitivity of plasma biomarkers, such as p-tau181, in tracking cognitive changes during treatment.
2. **Interpretation**: Lecanemab safely and effectively improved cognitive function in Chinese patients with early AD in a short-term real-world study. Treatment significantly reduced plasma p-tau181 and p-tau217 levels, with changes closely correlating with improvements in Alzheimer's Disease Assessment Scale-Cognitive Subscale 14-item version (ADAS-cog14 (ADAS-cog14) scores. Clinical benefits were observed across both sexes and apolipoprotein E4 (*APOE4)* carrier statuses. However, efficacy was limited and the risk of Amyloid-related imaging abnormalities (ARIA) increased in patients with moderate-to-severe AD.
3. **Future directions**: Given potential ethnic and regional differences in treatment response, future research should explore the use of plasma p-tau181 and other biomarkers for early identification of responders and dynamic monitoring of therapeutic effects, thereby supporting personalized treatment strategies across diverse populations.

2.4 | Statistical analyses

All statistical analyses and data visualization were performed using R software (version 4.2.3), SPSS (version 27.0), and GraphPad Prism (version 10.1.2). Statistical significance was defined as a two-tailed *p*-value < 0.05. The Shapiro–Wilk test was used to assess the normality of the variables. Continuous variables with normal distributions are presented as means and standard deviations (SDs), whereas those with non-normal distributions were presented as medians and interquartile ranges (IQRs). Categorical variables were presented as numbers and percentages.

First, a paired sample *t*-test or the Wilcoxon signed-rank test was used to compare the baseline and post-treatment scale scores and plasma biomarkers. Pearson or Spearman correlation analyses were conducted to assess the relationships between baseline clinical variables and cognitive scores or plasma biomarkers. Additionally, all cognitive scores and plasma biomarkers data collected during treatment were analyzed to examine correlations between cognitive performance and plasma biomarkers over time. Based on these analyses, plasma biomarkers significantly associated with cognitive scores were identified as candidate predictors. Second, patients were

classified as treatment-responsive or less-responsive depending on whether they demonstrated an improvement of ≥4 points in the ADAS-cog14 score from baseline (V0) to 7 months (V2). We then used area under the curve (AUC) to investigate whether baseline levels of these candidate predictors could predict therapeutic response to lecanemab after 7 months of treatment. A linear mixed-effects model, with sex and *APOE* genotype included as a covariate, was further employed to analyze the temporal changes in cognitive scores in dynamic relation to these plasma biomarkers. Thirdly, subgroup analyses were performed based on sex, *APOE4* carrier status, and CDR grading (≤1 or ≥2), to explore the effects of these factors on treatment efficacy.

3 | RESULTS

3.1 | Participants

The study flowchart was presented in Figure S1, and the baseline demographic characteristics of all participants were summarized in Table 1. A total of 68 patients from multiple centers were included, consisting of 50 females and 18 males. The mean age at diagnosis was 68.74 ± 11.20 years, with an average body mass index (BMI) of 21.03 ± 3.19 kg/m^2. The participants had a mean educational attainment of 11.75 ± 4.01 years, and a mean whole-brain SUVR of 1.43 ± 0.30. *APOE* genotyping revealed the following allele frequencies: *ε2/ε3* in 2.94%, *ε2/ε4* in 1.47%, *ε3/ε3* in 35.29%, *ε3/ε4* in 48.53%, and *ε4/ε4* in 11.76%. Based on CDR scores, 36.76% of patients were classified as AD-MCI (CDR = 0.5), 39.71% had mild dementia (CDR = 1), 22.06% had moderate dementia (CDR = 2), and 1.47% had severe dementia (CDR = 3). Pathogenic mutations were identified in three patients: one each in *PSEN1*, *PSEN2*, and *APP*. In terms of pharmacological management, 97.06% of patients were treated with cholinesterase inhibitors, and 17.65% received N-methyl-D-aspartate (NMDA) receptor antagonists (Table 1).

3.2 | Efficacy of Lecanemab treatment

Data from all treated patients were collected longitudinally at baseline (V0, $n = 68$), 2.5 months (V1, $n = 58$), and 7 months (V2, $n = 34$) (Table 2; Figure 1A–F). The ADAS-cog14 showed significant declines at both V1 (−4.32 points; 95% confidence interval [CI]: −7.72 to −0.91; $p = 0.004$) and V2 (-4.62 points; 95% CI: −6.74 to −2.50, $p < 0.001$). Similarly, the CDR-SB also decreased at V1 (−0.46; 95% CI: −0.77 to −0.45, $p = 0.002$) and V2 (−0.16; 95% CI: −0.77 to −0.04, $p = 0.038$). The MoCA improved significantly at V1 (+1.09; 95% CI: 0.18 to 2.00; $p = 0.020$) and V2 (+2.00; 95% CI: 0.67 to 3.33; $p = 0.004$), while the MMSE showed a mild increase at V2 (+1.26; 95% CI: 0.04 to 2.49; $p = 0.043$). In contrast, the FAB demonstrated a significant improvement only at V1 (+0.64; 95% CI: 0.02 to 1.27, $p = 0.044$).

TABLE 1 Baseline characteristics.

Variables	Value
Sex (female/male)	50/18
Age at diagnosis (years)	68.74 ± 11.20
BMI (kg/m^2)	21.03 ± 3.19
Years of education	11.75 ± 4.01
The whole-brain SUVR	1.43 ± 0.30
APOE gene	
ε2/ε3 (*n*, %)	(2, 2.94%)
ε2/ε4 (*n*, %)	(1. 1.47%)
ε3/ε3 (*n*, %)	(24. 35.29%)
ε3/ε4 (*n*, %)	(33. 48.53%)
ε4/ε4 (*n*, %)	(8. 11.76%)
CDR	
0.5 (*n*, %)	(25, 36.76%)
1 (*n*, %)	(27, 39.71%)
2 (*n*, %)	(15, 22.06%)
3 (*n*, %)	(1, 1.47%)
Pathogenic gene	
PSEN1	(1,1.47%)
PSEN2	(1,1.47%)
APP	(1,1.47%)
Drugs	
Cholinesterase inhibitors	(66, 97.06%)
NMDAR antagonist	(12, 17.65%)

Abbreviations: *APOE*, apolipoprotein E; BMI, body mass index; CDR, clinical dementia rating; SUVR, standardized uptake value ratio (Reference regions: whole cerebellum).

Among plasma biomarkers, p-tau181 levels significantly decreased at both V1 (−2.05 pg/ml; 95% CI: −3.06 to −1.04; $p < 0.001$) and V2 (−3.57 pg/ml; 95% CI: −5.10 to −2.04; $p < 0.001$; Figure 2B). Similarly, the p-tau181/Aβ1-42 ratio showed significant reductions at both V1 (-1.50; 95% CI: −5.09 to −0.08; $p = 0.020$) and V2 (−2.79; 95% CI: −6.23 to −0.66; $p = 0.014$; Figure 2C). Compared with V0, plasma p-tau217 did not show change at V1 (−0.10 pg/ml; 95% CI: −0.36 to 0.16; $p = 0.450$), but was significantly decreased at V2 (−0.27 pg/ml; 95% CI: −0.48 to −0.07; $p = 0.012$, Figure 3D). However, other plasma biomarkers, including the Aβ1-42/Aβ1-40 ratio, GFAP, and NfL, showed no statistically significant changes at either V1 or V2 compared with V0 (all $p > 0.05$; Figure 2E–G).

A total of 12 patients (17.65%) completed follow-up Aβ-PET/CT scans. The SUVR values for all brain regions and the whole-brain before and after treatment were summarized in Table S1. After about 1-year treatment, patients showed a significant reduction in mean whole-brain SUVR (−0.20; 95% CI: −0.31 to −0.10; $p = 0.001$; Figure 2G), and prominent decreases were also observed in all brain regions except the bilateral occipital cortex (Table S2).

TABLE 2 Patient characteristics at baseline, 2.5-month, and 7-month follow-up.

Variables	V0 (*n* = 68)	V1 (*n* = 58)	V2 (*n* = 34)	Normal reference range
Age at diagnosis (years)	68.74 ± 11.20	68.68 ± 10.96	66.97 ± 10.42	
Sex (female/male)	50/18	44/16	24/10	
APOE-ε4 carriers (*n*, %)	(41, 70.70%)	(33, 56.90%)	(19, 55.90%)	
ADAS-cog14	35.67 ± 11.11	32.46 ± 16.36	31.15 ± 15.24	
CDR-SB	5.67 ± 3.32	5.52 ± 3.32	5.80 ± 3.32	
MMSE	19.69 ± 10.91	18.60 ± 6.71	18.71 ± 6.33	
MoCA	15.86 ± 7.71	16.38 ± 7.06	16.76 ± 6.28	
FAB	11.26 ± 4.35	11.60 ± 4.40	11.44 ± 4.31	
Aβ 1-42	62.11 (42.60–101.41)	85.49 (37.98–129.40)	83.13 (37.98–118.56)	≥4.7 pg/ml
Aβ 1-40	3.16 (1.91–5.00)	4.16 (1.95–7.21)	3.57 (2.64–4.70)	
Aβ1-42/Aβ1-40	0.05 (0.04–0.06)	0.05 (0.04–0.07)	0.06 (0.04–0.08)	≥0.05
p-tau181	8.69 ± 3.54	7.58 ± 3.22	6.29 ± 3.99	≤5 pg/ml
p-tau217	1.67 ± 0.84	1.55 ± 1.02	1.19 ± 0.80	≤0.4 pg/ml
p-tau181/Aβ 1-42	5.29 ± 8.71	2.79 ± 2.63	2.25 ± 2.24	
GFAP	177.33 (122.75–237.58)	174.21 (136.15–239.90)	134.08 (54.07–172.63)	≤125 pg/ml
NfL	66.21 (41.98–93.24)	60.82 (46.20–82.62)	56.03 (33.84–96.01)	≤150 pg/ml

Abbreviations: ADAS-cog14, Alzheimer's Disease Assessment Scale–Cognitive Subscale (14-item version); *APOE*, apolipoprotein E; CDR-SB, Clinical Dementia Rating–Sum of Boxes; FAB, Frontal Assessment Battery; GFAP, glial fibrillary acidic protein;MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; NfL, neurofilament light chain.

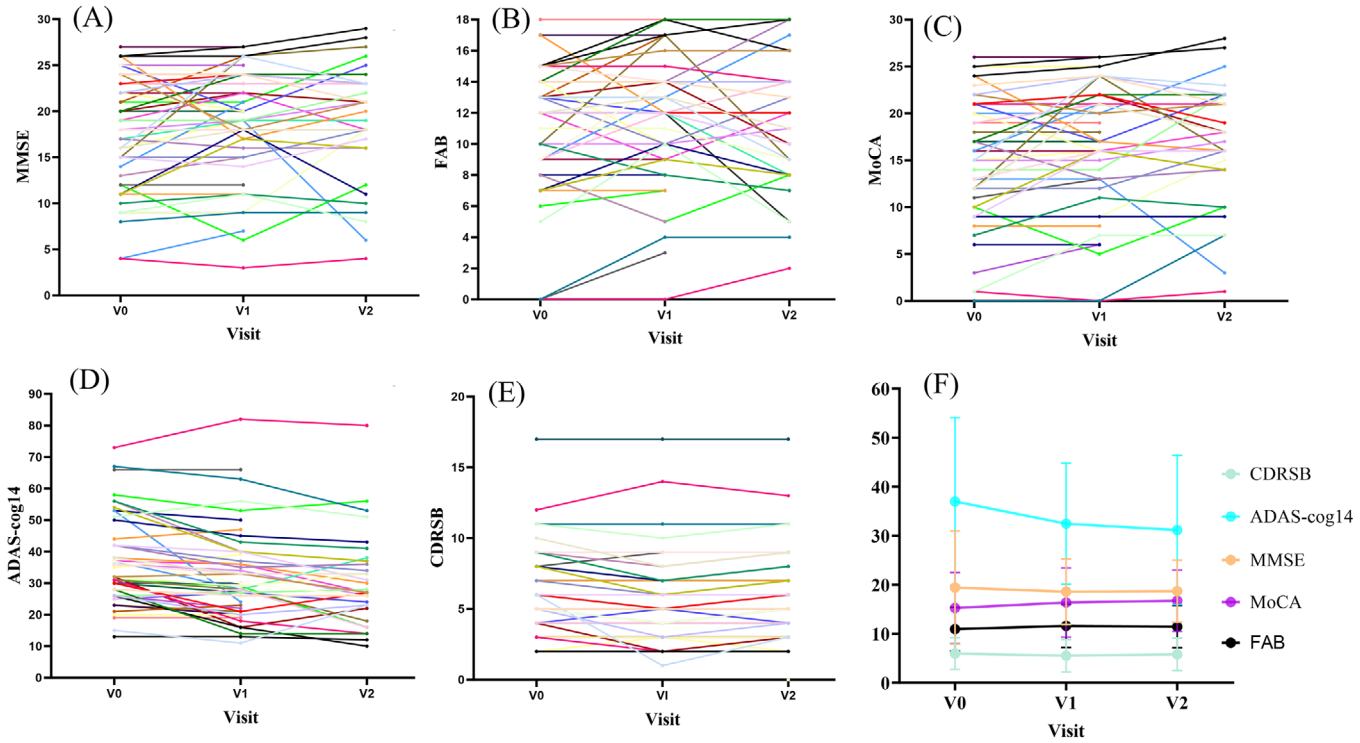

FIGURE 1 Cognitive scores before and after treatment. MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; FAB, Frontal Assessment Battery; CDR-SB, Clinical Dementia Rating-Sum of Boxes; ADAS-cog14, Alzheimer's Disease Assessment Scale-Cognitive Subscale 14-item version.

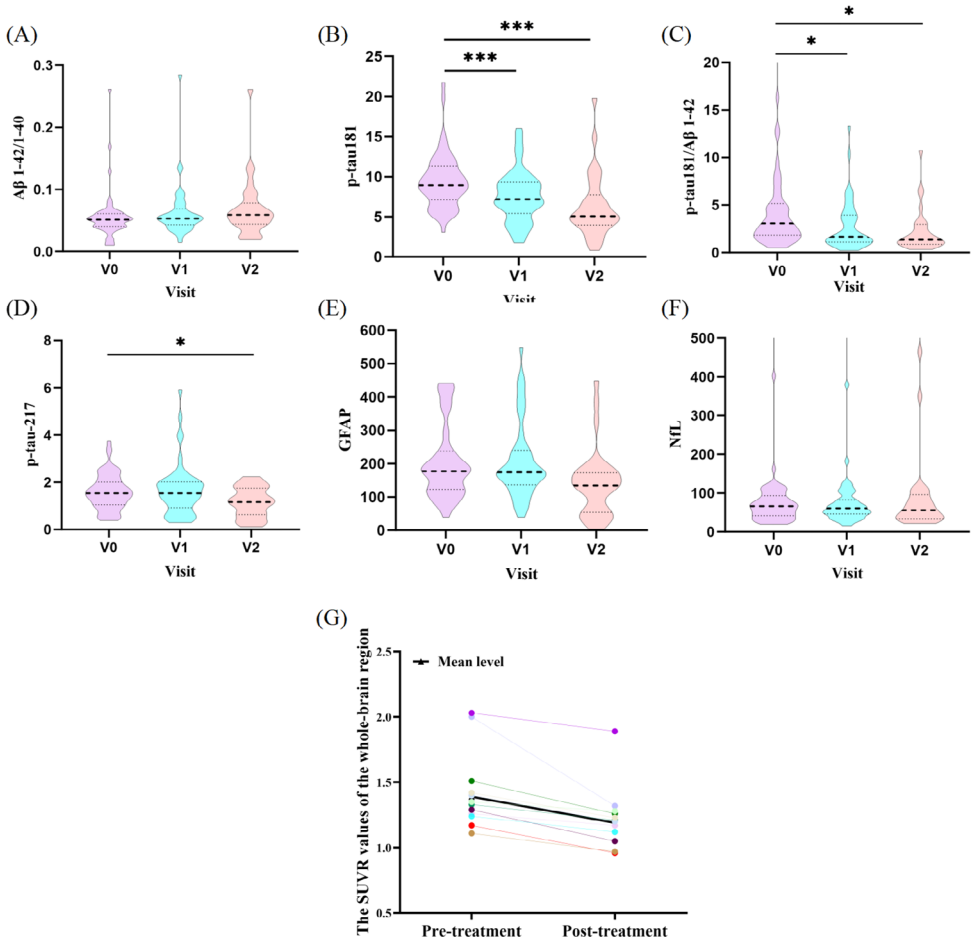

FIGURE 2 Plasma biomarkers and SUVR values pre- and post- treatment. GFAP, glial fibrillary acidic protein; NfL, neurofilament light chain. SUVR, standardized uptake value ratio.

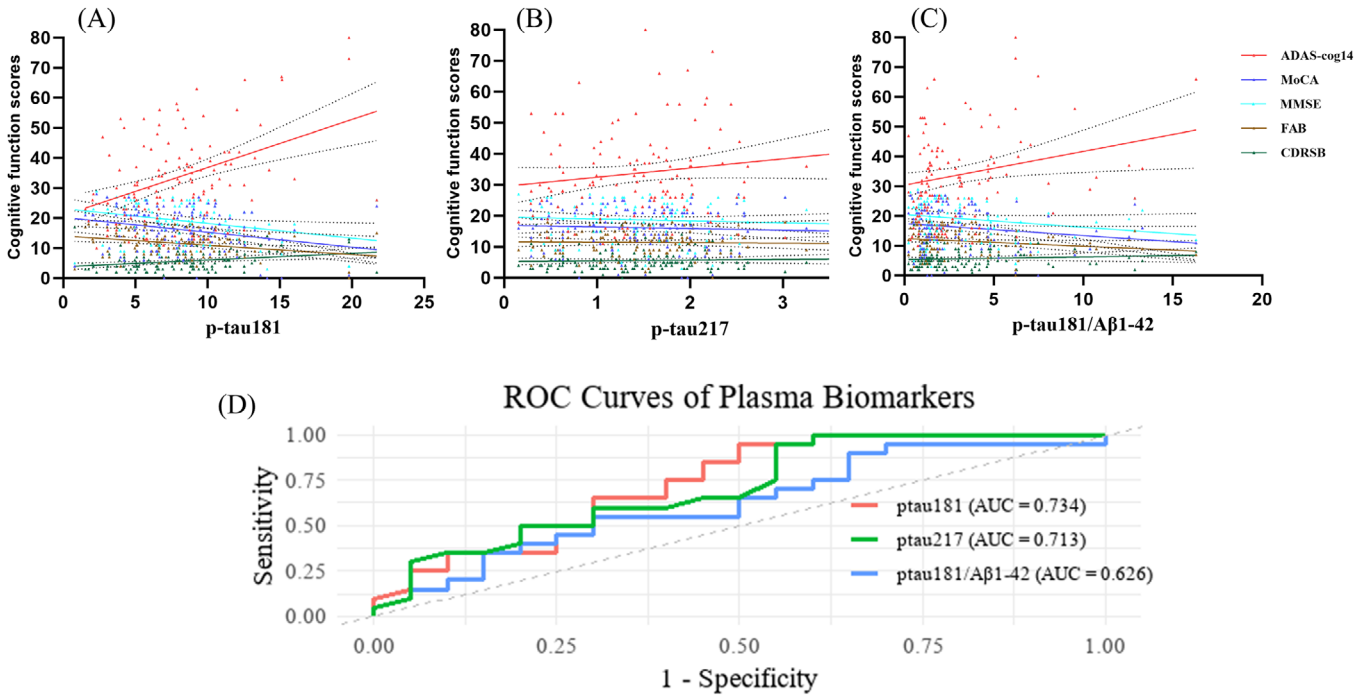

FIGURE 3 Correlation analysis of plasma biomarkers with cognitive scores and evaluation of their predictive value. AUC, area under the curve.

TABLE 3 Associations between cognitive assessments, plasma biomarkers, and clinical features at baseline.

Variables	Age at diagnosis [correlation coefficient/*p*)]	Sex	BMI	Years of education	*APOE* genotype
ADAS-cog14	(0.095/0.485)	(0.162/0.232)	(−0.064/0.639)	(−0.154/0.259)	(0.144/0.290)
CDR	(0.063/0.641)	(**0.282/0.021**)	(−0.083/0.310)	(−0.084/0.531)	(0.111/0.408)
CDR-SB	(0.006/0.967)	(0.183/0.178)	(−0.169/0.214)	(−0.042/0.756)	(0.144/0.291)
MMSE	(−0.119/0.384)	(−0.262/0.051)	(0.187/0.168)	(0.231/0.086)	(−0.102/0.453)
MoCA	(0.005/0.969)	(−**0.241/0.050**)	(0.230/0.089)	(0.126/0.354)	(−0.145/0.286)
FAB	(−0.136/0.319)	(−0.126/0.356)	(0.126/0.354)	(0.178/0.189)	(−0.090/0.508)
Aβ1-40	(0.134/0.324)	(0.003/0.985)	(−0.111/0.415)	(0.049/0.720)	(−0.074/0.590)
Aβ1-42	(0.037/0.787)	(0.133/0.330)	(−0.095/0.484)	(−0.046/0.739)	(0.184/0.174)
Aβ1-42/Aβ1-40	(−0.115/0.401)	(0.073/0.594)	(0.221/0.102)	(−0.120/0.379)	(0.266/0.048)
p-tau181	(−0.202/0.136)	(−0.089/0.513)	(0.102/0.453)	(0.151/0.266)	(0.196/0.148)
p-tau217	(−0.153/0.229)	(−0.062/0.625)	(−0.007/0.961)	(−0.112 /0.376)	(**0.247/0.049**)
p-tau181/Aβ1-42	(−0.111/0.413)	(−0.135/0.320)	(0.100/0.463)	(0.099/0.469)	(−0.103/0.450)
GFAP	(0.172/0.204)	(0.237/0.078)	(−0.025/0.857)	(0.127/0.352)	(0.026/0.849)
NfL	(0.212/0.117)	(0.158/0.244)	(−0.221/0.102)	(−0.069/0.613)	(0.010/0.941)

Abbreviations: ADAS-cog14, Alzheimer's Disease Assessment Scale–Cognitive Subscale (14-item version); *APOE*, apolipoprotein E; BMI, body mass index; CDR-SB, Clinical Dementia Rating–Sum of Boxes; FAB, Frontal Assessment Battery; GFAP, glial fibrillary acidic protein; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; NfL, neurofilament light chain.

3.3 | Correlation analysis

Correlations between cognitive scores, clinical variables, and plasma biomarkers at baseline were summarized in Table 3. Baseline CDR stage ($\rho = 0.282$, $p = 0.021$) and MoCA score ($\rho = -0.241$, $p = 0.050$) were significantly associated with sex, and plasma p-tau217 levels were positively correlated with *APOE* genotype ($\rho = 0.247$, $p = 0.049$). In contrast, other demographic variables, including age at diagnosis, BMI, and years of education, showed no significant correlation with baseline cognitive scores or plasma biomarkers (Table 3). Baseline correlations between plasma biomarkers and cognitive scores were shown in Table S3. Plasma p-tau181, p-tau 181/Aβ1-42, p-tau217, Aβ1-42/1-40, and GFAP showed associations with cognition.

All cognitive scores and plasma biomarker data were included in correlation analyses to explore associations between cognitive performance and plasma biomarkers. Plasma p-tau181 and p-tau217 were significantly associated with multiple cognitive scores (Table S4, Figure 3A–C). Specifically, p-tau181 showed positive correlations with ADAS-cog14 ($r = 0.376$, $p < 0.001$) and negative correlations with MMSE ($r = -0.207$, $p = 0.018$), MoCA ($r = -0.268$, $p = 0.002$), and FAB ($r = -0.273$, $p = 0.002$, Figure 3A). Plasma p-tau217 also correlated positively with ADAS-cog14 ($r = 0.361$, $p < 0.001$) and negatively with MMSE ($r = -0.264$, $p = 0.002$), MoCA ($r = -0.293$, $p < 0.001$), and FAB ($r = -0.285$, $p < 0.001$, Figure 3B). The p-tau181/Aβ1-42 showed modest correlation with ADAS-cog14 ($r = 0.228$, $p = 0.011$) and CDR-SB ($r = 0.175$, $p = 0.047$, Figure 3C). In contrast, GFAP, NfL, and Aβ1-42/Aβ1-40 showed no significant correlations with most cognitive scores (all $p > 0.05$, Table S4).

At V2, 51.28% of patients demonstrated a ≥4-point reduction in ADAS-cog14 scores compared to V0, and were classified as the treatment-responsive group. We evaluated the predictive value of baseline plasma biomarkers, including p-tau181, p-tau217, and the p-tau181/Aβ1-42 ratio, which were significantly associated with cognitive scores. We found that these biomarkers showed significant changes following treatment, and can be used to identify responders after 7 months of lecanemab treatment. Baseline plasma p-tau181 showed the highest predictive performance (AUC = 0.734) with an optimal cutoff of 11.82 pg/ml (sensitivity 95.2%, specificity 47.4%; Figure 3D), followed by p-tau217 (AUC = 0.713; cutoff 1.51 pg/ml; sensitivity 72.5%, specificity 68.3%) and p-tau181/Aβ1-42 (AUC = 0.626; cutoff 4.32; sensitivity 63.8%, specificity 65.5%, Figure 3D).

3.4 | Longitudinal association analysis

To further investigate the longitudinal associations between plasma biomarkers and cognitive outcomes, we employed linear mixed-effects models incorporating time as an interaction term and adjusting covariates. First, after adjusting for sex, plasma p-tau181 levels were significantly associated with changes in ADAS-cog14 scores, with higher levels indicating worse cognitive performance ($\beta = 1.34$, 95% CI: 1.04 to 1.75, $p = 0.026$; Table 4). No significant associations were observed between p-tau181 and CDRSB, MoCA, MMSE, or FAB scores. Longitudinal analyses also found that the p-tau181/Aβ1-42 ratio ($\beta = -0.58$, 95% CI: −1.15 to −0.11, $p = 0.017$) was significantly associated with MMSE scores after adjusting for sex. Since baseline p-tau217 was

TABLE 4 Temporal correlation between plasma biomarkers and cognitive scores.

Variables	ADAS-cog14		CDR-SB		MoCA		MMSE		FAB	
	β (95% CI)	*p*-value	β (95% CI)	*p*-value	β (95% CI)	*p*-value	β (95% CI)	*p*-value	β (95% CI)	*p*-value
p-tau181[a]	1.340 (1.040, 1.750)	**0.026**	0.03 (−0.060, 0.130)	0.489	−0.300 (−0.660, 0.050)	0.102	−0.240 (−0.600, 0.110)	0.182	−0.150 (−0.510, 0.110)	0.252
p-tau181/Aβ1-42[a]	−0.294 (−1.317, 0.729)	0.575	0.03 (−0.060, 0.130)	0.489	−0.41 (−0.900, −0.080)	0.104	−0.580 (−1.150, −0.110)	**0.017**	−0.08 (−0.450, 0.280)	0.667
p-tau217[a]	2.345 (−0.508, 5.198)	0.110	0.026 (−0.512, 0.564)	0.924	−0.784 (−2.060, 0.490)	0.230	−0.370 (−1.512, 0.772)	0.527	−0.113 (−1.058, 0.833)	0.816
p-tau181[b]	−0.302 (−1.062, 0.458)	0.438	−0.066 (−0.150, 0.017)	0.123	−0.148 (−0.395, 0.101)	0.247	−0.500 (−1.059, 0.059)	0.082	−0.018 (−0.174, 0.138)	0.824
p-tau181/Aβ1-42[b]	0.303 (−1.320, 0.715)	0.561	−0.126 (−0.536, 0.284)	0.548	−0.802 (−2.092, 0.488)	0.226	−0.363 (−1.518, 0.791)	0.539	−0.150 (−0.936, 0.637)	0.710
p-tau217[b]	2.280 (0.599, 5.160)	**0.020**	−0.075 (−0.182, 0.331)	0.180	−1.567 (−3.204, 0.071)	0.063	−1.215 (−2.673, 0.643)	0.105	−0.100 (−1.058, 0.858)	0.838

Abbreviations: ADAS-cog14, Alzheimer's Disease Assessment Scale–Cognitive Subscale (14-item version); *APOE*, apolipoprotein E; CDR-SB, Clinical Dementia Rating–Sum of Boxes; CI, confidence interval; FAB, Frontal Assessment Battery; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment.

[a]Covariates: sex.

[b]Covariates: sex and *APOE* type.

associated with *APOE* genotype (Table 3), we further adjusted for both sex and *APOE* genotype, revealing a significant correlation between plasma p-tau217 levels and ADAS-cog14 scores (β = 2.28, 95% CI: 0.60–5.16, $p = 0.020$; Table 4).

3.5 | Subgroup analysis

3.5.1 | Different sexes

Among male patients, ADAS-cog14 scores significantly declined at both V1 (−6.79 points; 95% CI: −10.04 to −3.53, $p < 0.001$) and V2 (−4.99 points; 95% CI: −7.10 to −3.74 $p = 0.001$). CDR-SB scores were also significantly reduced at V1 (−0.64; 95% CI: −1.13 to −0.16; $p = 0.013$), whereas MMSE, MoCA, and FAB scores remained stable throughout the study. At V1, no plasma biomarkers showed significant changes. At V1, plasma biomarkers showed no significant changes. By V2, p-tau181 (−5.79 pg/ml; 95% CI: −11.07 to −2.76; $p = 0.036$) and p-tau217 (-0.47 pg/ml; 95% CI: −0.76 to −0.10) decreased significantly (Table S5). In female patients, ADAS-cog14 scores significantly decreased at V2 (-4.27; 95% CI: −5.00 to −2.84; $p = 0.045$). Plasma p-tau181 levels declined at both V1 (-1.83 pg/ml; 95% CI: −2.59 to −1.08; $p = 0.003$) and V2 (−4.07 pg/ml; 95% CI: −6.07 to −2.06; $p < 0.001$), while p-tau217 decreased significantly at V2 (-0.30 pg/ml; 95% CI: −0.52 to −0.09; $p = 0.009$; Table S6).

3.5.2 | *APOE ε4* carrier status

In *APOE ε4* carriers, ADAS-cog14 scores improved at V1 (−4.17; 95% CI: −6.75 to −1.58; $p = 0.003$) and V2 (-5.10; 95% CI: −7.17 to −2.54; $p < 0.001$). CDR-SB also decreased at V1 (-0.48; 95% CI: −0.77 to −0.20; $p = 0.001$) and V2 (-0.15; 95% CI: −0.28 to −0.02; $p = 0.023$), while MoCA increased at both visits (+1.20 and +2.72; both $p < 0.05$). MMSE and FAB remained unchanged. Plasma p-tau181 declined at V1 and V2, and p-tau217 at V2 (all $p < 0.05$; Table S7). Similar trends were observed in non-*APOEε4* carriers, with ADAS-cog14 and p-tau181 decreasing at both visits and p-tau217 at V2 (all $p < 0.05$; Table S8).

3.5.3 | CDR staging

In patients with AD-MCI and mild dementia (CDR ≤ 1), ADAS-cog14 scores significantly declined at V1(-2.88; 95% CI: −4.43 to −1.34; $p < 0.001$) and V2 (-5.19; 95% CI: −7.33 to −3.04; $p < 0.001$, Table S8), accompanied by reductions in CDR-SB scores ($p = 0.005$ and $p = 0.018$, respectively). MoCA scores increased modestly at V2 ($p = 0.039$), while MMSE and FAB scores remained unchanged (all $p > 0.05$). Among plasma biomarkers, p-tau181 levels decreased significantly at both follow-up visits, while p-tau217 declined significantly at V2 (Table S9, all $p < 0.05$). In moderate-to-severe AD patients, we only found MMSE scores significantly increased at V1 (+1.71; 95% CI: −3.15 to −0.27; $p = 0.023$), and MoCA scores also improved at V1 (+2.71; 95% CI:

TABLE 5 Adverse reactions and outcomes in patients.

Type	Cases (*n*, %)	Severity
Infusion reaction	(4, 5.88%)	Mild; one patient received symptomatic treatment with medication
Pyrexia (*n* = 4)	(4, 5.88%)	
Headache (*n* = 3)	(3, 4.41%)	
Arthralgia (*n* = 1)	(1, 1.47%)	
Fatigue (*n* = 2)	(2, 2.94%)	
Nausea (*n* = 1)	(1, 1.47%)	
ARIA	(12, 17.65%)	
ARIA-E ± H	(9, 13.24%)	Asymptomatic
Isolated ARIA-H	(3, 4.41%)	
Other		
Psychiatric symptoms	(1, 1.47%)	Moderate
Discontinue treatment	(4, 5.88%)	
ARIA	(2, 2.94%)	
Psychiatric symptoms	(1, 1.47%)	
Traumatic intracranial hemorrhage	(1, 1.72%)	

Abbreviations: ARIA-E, amyloid-related imaging abnormalities-edema; ARIA-H, amyloid-related imaging abnormalities-hemosiderin deposition.

−4.34 to −1.09; $p = 0.003$). Among plasma biomarkers, both p-tau181 and GFAP exhibited a downward trend, but the changes did not reach statistical significance (Table S10).

3.6 | Individuals with pathogenic gene

Cognitive scores and plasma biomarker levels showed fluctuations with no substantial overall improvement (Figure S2, Tables S11 and S12). Notably, the patient carrying the *PSEN2* developed asymptomatic ARIA after 7 months of treatment.

3.7 | Safety

Lecanemab demonstrated a favorable overall safety profile, with an incidence of infusion-related reactions of 5.88% (Table 5). All infusion-related events were mild in severity and occurred after the first two infusions; only one patient required symptomatic treatment. In the overall cohort, ARIA occurred in 12 patients (17.65%), including nine cases of ARIA-E (with or without ARIA-H) and three cases of isolated ARIA-H. Among moderate-to-severe AD patients, the incidence of ARIA was as high as 43.75% (7/16), but in those with AD-MCI and mild AD, it was 9.62% (5/52). All 12 patients were asymptomatic, and ARIA was incidentally detected by MRI prior to the fifth infusion ($n = 2$) or the seventh infusion ($n = 10$). Notably, only two patients showed more than four cerebral microhemorrhages on MRI, both of whom had moderate AD (CDR = 2). One patient with pre-existing severe white matter lesions (Fazekas III, CDR = 2) developed psychiatric symptoms during treatment.

4 | DISCUSSION

In this real-world, multicenter study conducted in China, we report several key findings: (1) lecanemab demonstrated clinical efficacy in patients with AD-MCI and mild AD, as evidenced by improvements in cognitive function following both the fifth and seventh infusions, alongside significant reductions in plasma p-tau181 and p-tau217 levels. (2) Both male and female patients, as well as *APOE4*-carriers and non-carriers, could be benefit from treatment; however, disease severity was a major determinant of response—patients with more advanced disease showed less favorable outcomes. (3) Baseline plasma p-tau181 and p-tau217 levels may predict treatment response, and longitudinal examination of these biomarkers during therapy could provide valuable insights into therapeutic efficacy and inform clinical decision-making. (4) Lecanemab was generally well tolerated, with a low and manageable incidence of infusion-related reactions and ARIA, supporting its safety in a Chinese real-world clinical context.

Compared to participants in clinical trials, patients in our real-world cohort exhibited more severe cognitive impairment at baseline.[12,14] Nevertheless, early improvements in cognitive performance and reductions in plasma p-tau181 levels were observed following lecanemab treatment. Given the current scarcity of real-world evidence on lecanemab, and the predominant focus of existing studies on safety outcomes, our findings provide important preliminary insights into both efficacy and biomarker response.[16–20] Several factors may have contributed to the observed therapeutic effects. A U.S. real-world study reported an estimated annual CDR-SB progression rate of 1.11 in early-stage AD patients treated with lecanemab, slightly better than the 1.21 rate in the Clarity-AD, despite worse baseline cognition function.[12,16] Moreover, Asian patients in the Clarity-AD exhibited higher rates of amyloid clearance and greater reductions in amyloid burden, indicating a potentially enhanced responsiveness to lecanemab in this population, despite the absence of data from Chinese cohorts.[14] Furthermore, in the Clarity-AD clinical trial, 61.5% of participants were diagnosed with AD-MCI, a stage at which the potential for measurable cognitive improvement was relatively limited, and cognitive outcome data for the overall study population were not reported.[12] Additionally, 67.24% of our patients were newly diagnosed and initiated treatment with cholinesterase inhibitors or NMDA receptor antagonists concurrently with lecanemab. These oral medications may have contributed synergistically to the early cognitive improvements observed. This concurrent use is common in real-world practice and was also permitted in the Clarity-AD trial.[17] However, beyond symptomatic benefits, we observed a significant early decline in plasma p-tau181, a biomarker closely associated with AD pathology, indicating a potential disease-modifying effect of lecanemab.[28–30]

Our results showed that plasma p-tau181 and p-tau217 levels both declined significantly after treatment, with p-tau181 appearing to decrease earlier. Plasma p-tau181 levels declined in parallel with improvements in cognitive function following lecanemab treatment, exhibiting a particularly strong dynamic correlation with changes in ADAS-cog14 scores. This association was stronger than those

observed with other plasma biomarkers. Notably, baseline p-tau181 levels showed no significant association with demographic variables such as age, sex, education level, or *APOE* genotype, underscoring its stability and potential applicability across diverse patient subgroups. This is in line with previous studies identifying plasma p-tau181 as a robust biomarker of cerebral AD pathology and cognitive decline.[30,31] These findings suggest that p-tau181 measured by the cost-effective chemiluminescence assay may serve as a more sensitive and reliable marker for monitoring treatment response, supporting its future clinical implementation. Moreover, we observed a significant reduction in plasma p-tau217 levels after 7 months of treatment, consistent with the findings from the TRAILBLAZER-ALZ 2 trial. These results suggest that lecanemab not only effectively clears cerebral Aβ deposits but also ameliorates the downstream tau pathology associated with amyloid burden.[32,33] Notably, in our study, the decline in p-tau181 preceded that of p-tau217 and showed a stronger correlation with cognitive improvement. In Study 201 and its open-label extension (OLE), where fluctuations in p-tau181 occurred earlier than changes detected by Aβ-PET and rebounded more rapidly after treatment discontinuation, underscoring its sensitivity as an early biomarker.[28] We also found that plasma p-tau181 demonstrated the highest sensitivity in predicting AD patients who achieved the MCID. These differences may be explained partly by their distinct biological characteristics. P-tau217 predominantly accumulates around established amyloid plaques and reflects plaque formation, whereas p-tau181 is more directly linked to axonal injury induced by Aβ burden.[34,35] However, we did not observe significant changes in plasma biomarkers that have been reported in other AD studies, including GFAP, NfL, and the Aβ1-42/Aβ1-40 ratio.[36–39] In real-world studies, additional confounding factors may exist, potentially leading to results that differ from those observed in randomized controlled trials, including the relatively small sample size and more severe condition of enrolled patients.[40,41]

We found baseline plasma p-tau181 demonstrated the highest predictive value for treatment response at 7 months compared other plasma indicators, with the AUC of 0.734, indicating moderate accuracy. The optimal cutoff of 11.82 pg/ml showed high sensitivity (95.2%), suggesting it may be a reliable indicator for identifying responders. However, the relatively low specificity (47.4%) indicates a high false-positive rate, suggesting that patients included below the threshold may not have derived a definitive therapeutic benefit. Conversely, although the AUC of plasma p-tau217 was 0.713, slightly lower than that of p-tau181, its sensitivity and specificity were relatively balanced at 72.5% and 68.3%, respectively, suggesting that it may serve as a more scientifically predictive biomarker. These findings also highlight the need for integrating additional indicators to improve the prediction of treatment response to lecanemab.

In subgroup analyses, we found that patients of different sexes and *APOE* ε4 carrier statuses benefited from lecanemab treatment, consistent with the Charity-AD.[12] Notably, greater cognitive improvement was observed in male patients compared to females. While this may suggest a true sex-related difference in treatment response, it could also be confounded by lower baseline cognitive performance among female participants and the smaller sample size of male patients, which may introduce statistical bias.[42] Moreover, baseline disease severity appeared to influence the efficacy of lecanemab. Among our patients with moderate-to-severe AD (CDR $\geq$ 2), both ADAS-cog14 scores and plasma p-tau181 levels showed a downward trend after 2.5 months of treatment, but these changes did not reach statistical significance. Improvements in MoCA and MMSE scores in this group may be partly attributable to practice effects from repeated short-interval testing.[43] Importantly, all three patients who discontinued treatment due to moderate ARIA or neuropsychiatric symptoms were moderate AD (CDR = 2), consistent with findings from the previous U.S. cohort.[16] This suggests that disease severity may be a risk factor for ARIA and highlights the need for cautious consideration when initiating lecanemab in patients with more advanced disease.[44] In line with the stance of Israeli investigators,[20] we do not plan to broaden the application of lecanemab in patients with moderate-to-severe AD either, given the reduced clinical benefits, increased risk of adverse events, and higher treatment burden observed in this subgroup.

In terms of safety, lecanemab demonstrated a favorable tolerability profile in our study cohort. On the one hand, the infusion-related reactions were significantly lower than that reported in the Asian population from the Charity-AD (5.88% vs. 12.3%).[14] Notably, no prophylactic medications such as diphenhydramine were administered, and all reactions were mild. Only one patient experienced transient high fever (39.2°C) after the first infusion, which resolved with low-dose corticosteroids and did not recur. On the other hand, the overall incidence of ARIA (including ARIA-E [edema/effusion] and ARIA-H [hemorrhage]) in our cohort was slightly higher than that reported in Asian clinical trial populations, but this difference may be attributable to the inclusion of patients with moderate-to-severe AD in our real-world study.[14] Among patients with AD-MCI and mild AD, the ARIA incidence in our cohort was only 9.62% (5/52), and all cases were asymptomatic, further supporting the safety of lecanemab in early-stage Chinese patients.

5 | LIMITATION

This study has several limitations. First, the relatively small sample size, short follow-up duration, and lack of a control group are common challenges in most real-world studies. Second, our study differs from clinical trials in several key aspects, such as the severity of patients' disease and the scales used for cognitive assessment and the methods for sample examination. Third, the practical implementation of comprehensive neuropsychological evaluations is often limited by factors such as time constraints, resource availability, and patient compliance.

6 | CONCLUSION

Lecanemab appears to be safe and effective in individuals with early AD in the real-world Chinese population, with low and manageable incidences of ARIA and infusion-related reactions. Patients of different sexes and *APOE4* carrier statuses may benefit from treatment, whereas caution is warranted in those with moderate-to-severe AD or

extensive white matter hyperintensities. Changes in plasma p-tau181 and p-tau217 levels were closely associated with changes in ADAS-cog14 scores during treatment, suggesting that dynamic examinations of these two biomarkers may be useful for tracking cognitive decline and guiding clinical decision-making.

AUTHOR CONTRIBUTIONS

Sihui Chen and Ruwei Ou conceived the study, performed data analysis, and drafted the manuscript. Qianqian Wei, Chunyu Li, Wei Song, Bi Zhao, Jing Yang, Jiajia Fu, Yuanzheng Ma, Jiyong Liu, Xiangming Wang, Dengfu Fang, Tao Hu, Li Xiao, Shushan Zhang, Rui Huang, Xiaoyan Guo, and Fei Feng contributed to the collection of clinical and plasma data. Xueping Chen and Huifang Shang contributed to the review, editing, and scientific research thinking and methods. All authors have read and approved the manuscript.

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CONFLICT OF INTEREST STATEMENT

We declare that there is no actual or potential financial and other conflict of interest related to the submitted manuscript. Author disclosures are available in the Supporting Information.

CONSENT STATEMENT

All authors approved its publication.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by the West China Hospital of Sichuan University Ethics Committee (No. 727). All participants gave their written informed consent to participate in the study.

DATA AVALIABILITY STATEMENT

Detailed data have been shown in the Supplementary Materials.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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